CGI 2025

Reproducing the Appearance of Metallic Materials by Capturing Long-Range Dependencies using Image-to-Image Translation Network with Vision Transformer

Kaito Kojima, Taishi Iriyama, Takashi Komuro
Saitama University

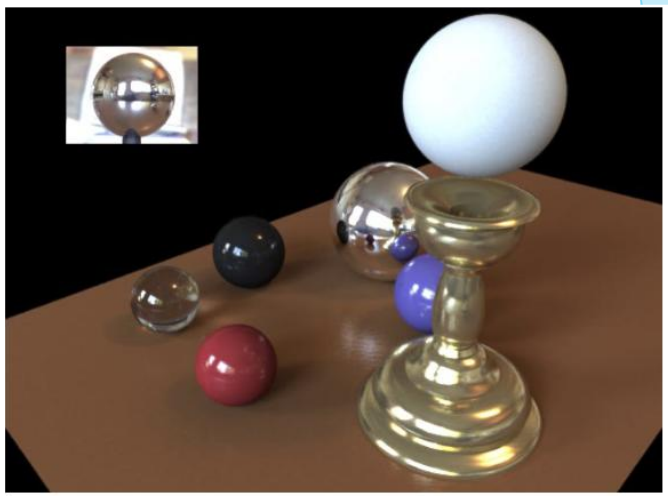
Presenter Kaito Kojima

Introduction

- In augmented reality (AR), reproducing natural shadows and reflections based on the lighting information of the real environment enhances the user's sense of immersion.

- In particular, when a virtual object is metallic, it is important to accurately reproduce the reflection of the surrounding environment on its surface.

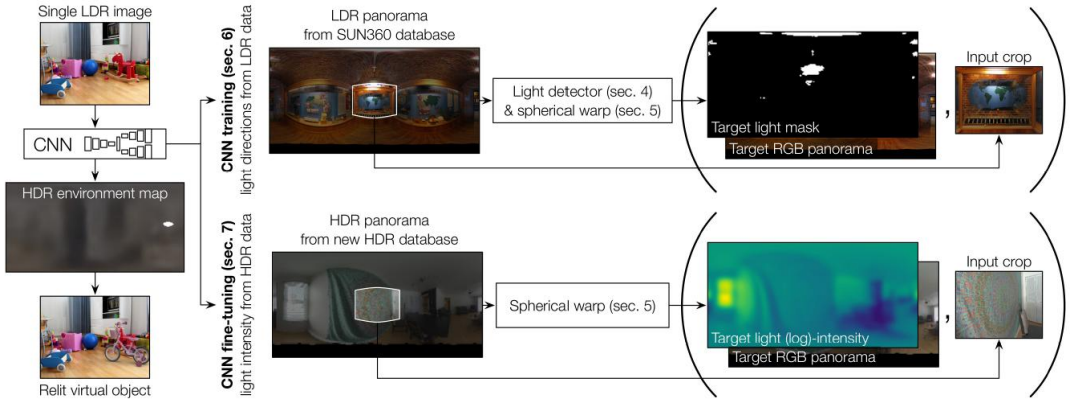
- However, preparing and setting up specialized equipment such as a mirror ball in the real world to capture the lighting information is time-consuming.



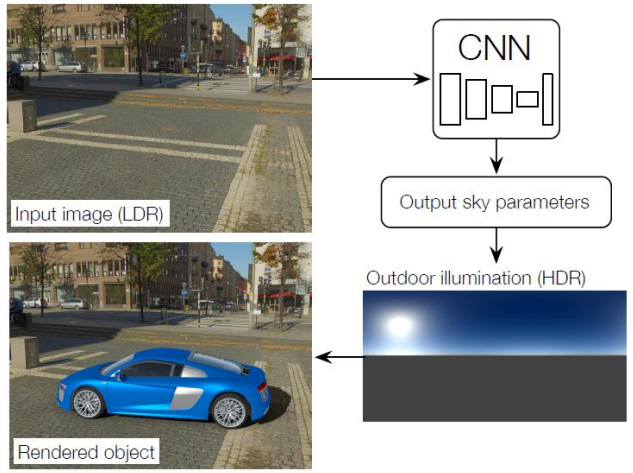
[Debevec, 1998]

Related Work (1/3)

Lighting estimation and generating shadows and reflections



[Gardner et al., 2017]

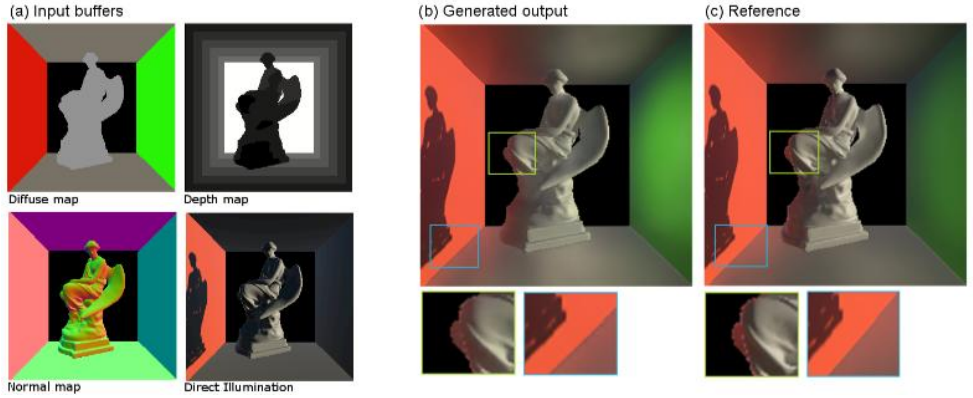


[Hold-Geoffroy et al., 2017]

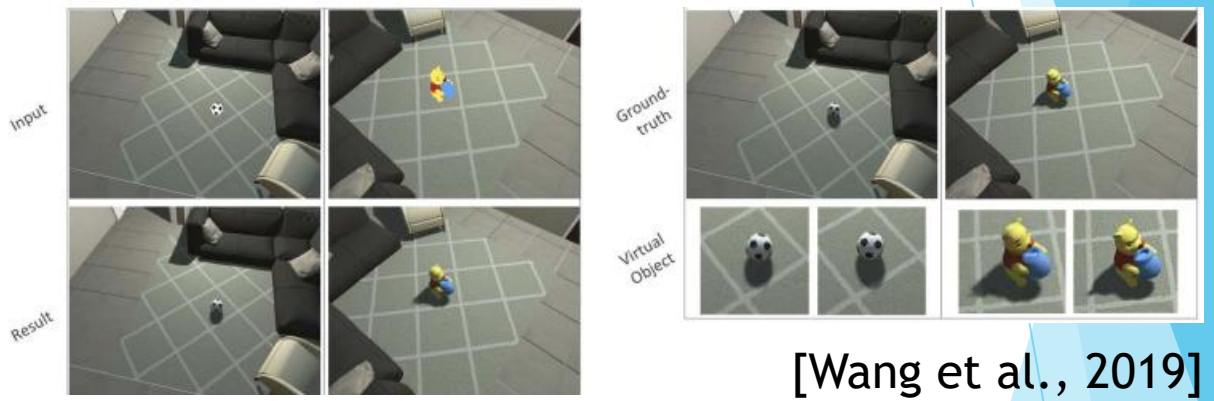
- No specialized equipment required.
- Reflections on metallic objects lack sharpness because the lighting is estimated using low-dimensional parameters.

Related Work (2/3)

Generating shadows and reflections using CNN-based image-to-image translation networks



[Thomas et al., 2017]

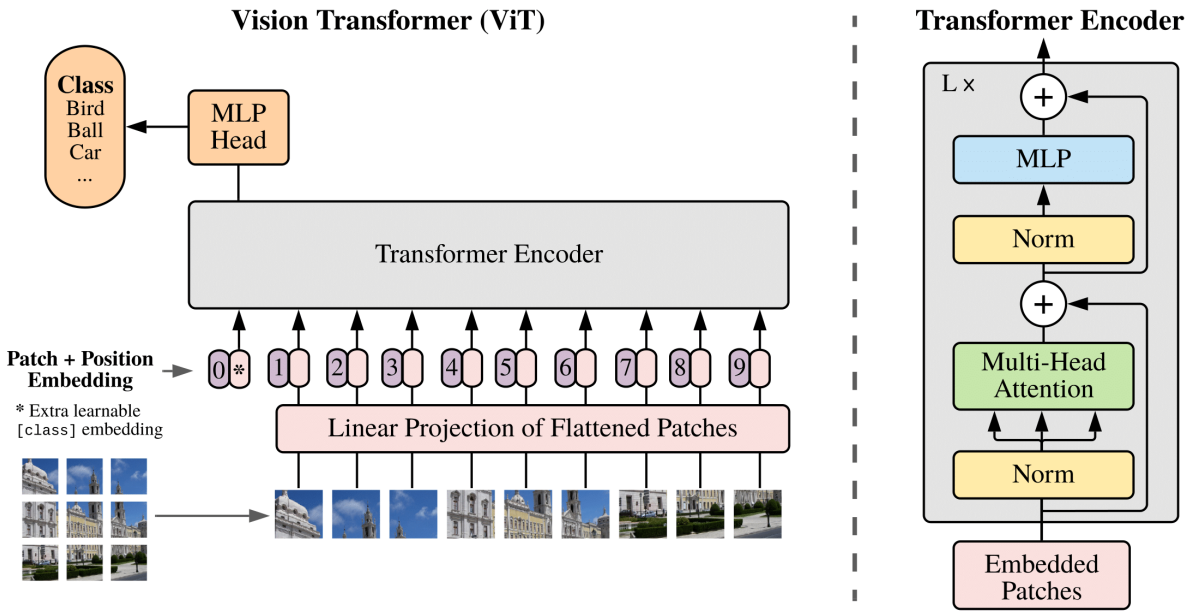


[Wang et al., 2019]

- These methods generate high-quality images that are consistently illuminated without intermediate steps.
- CNN-based models are less effective at capturing the long-range dependencies.

Related Work (3/3)

Vision Transformer (ViT)

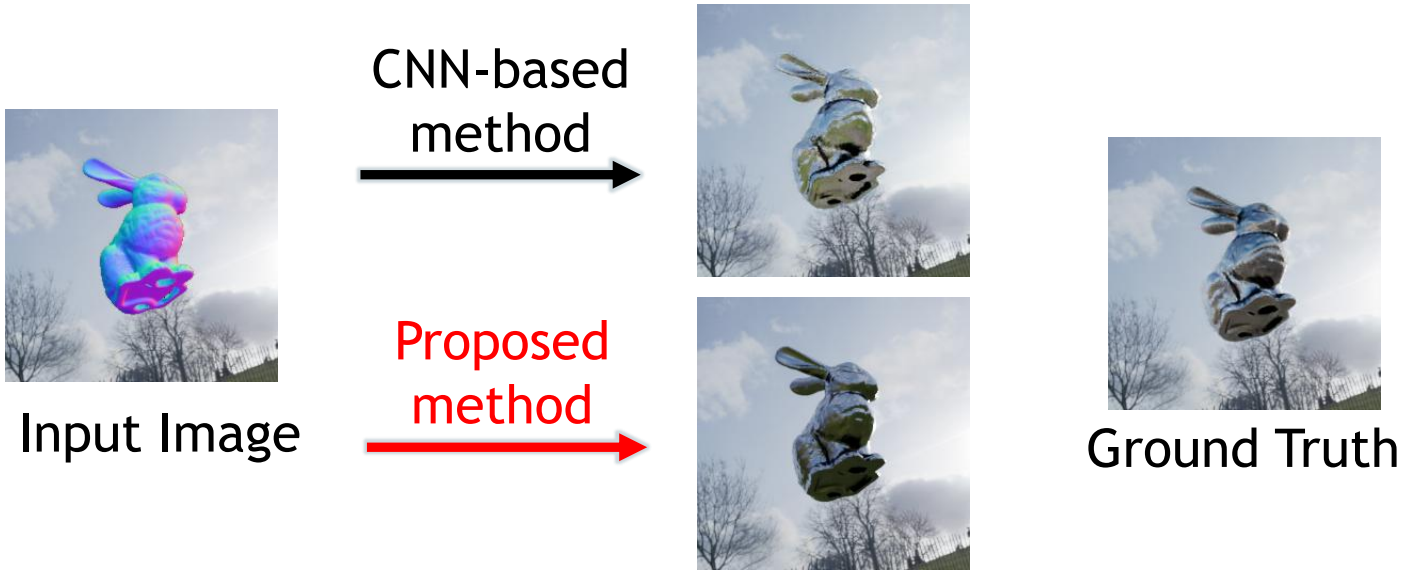


[Dosovitskiy et al., 2021]

- A new architecture that divides input images into small patches and captures the relationships between patches.
- ViT extracts global features from images, which was difficult to achieve using CNNs.

Purpose of the Study

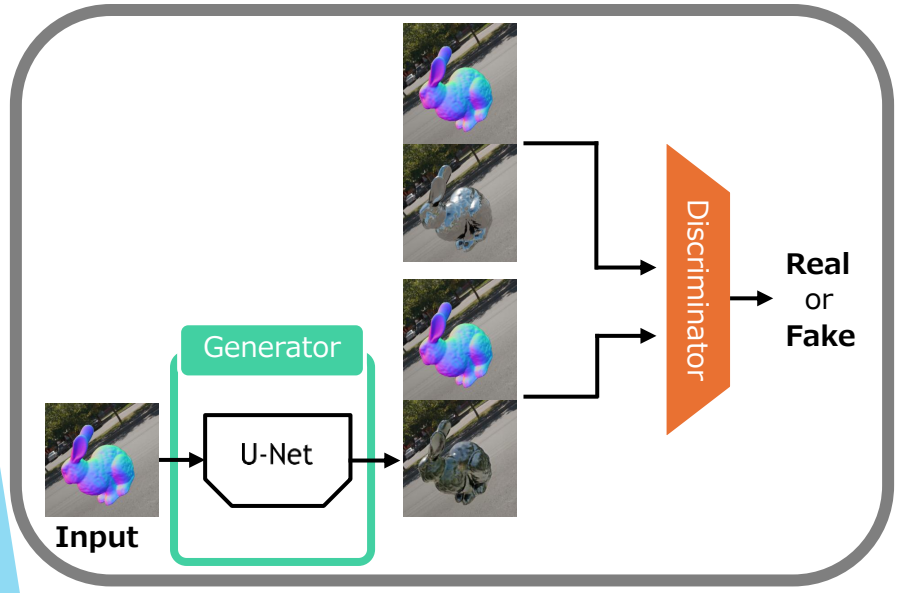
- To reproduce the appearance of metallic materials from a single image by using an image-to-image translation network
- CNN-based method cannot effectively capture surrounding lighting information, resulting in causing visual inconsistency.



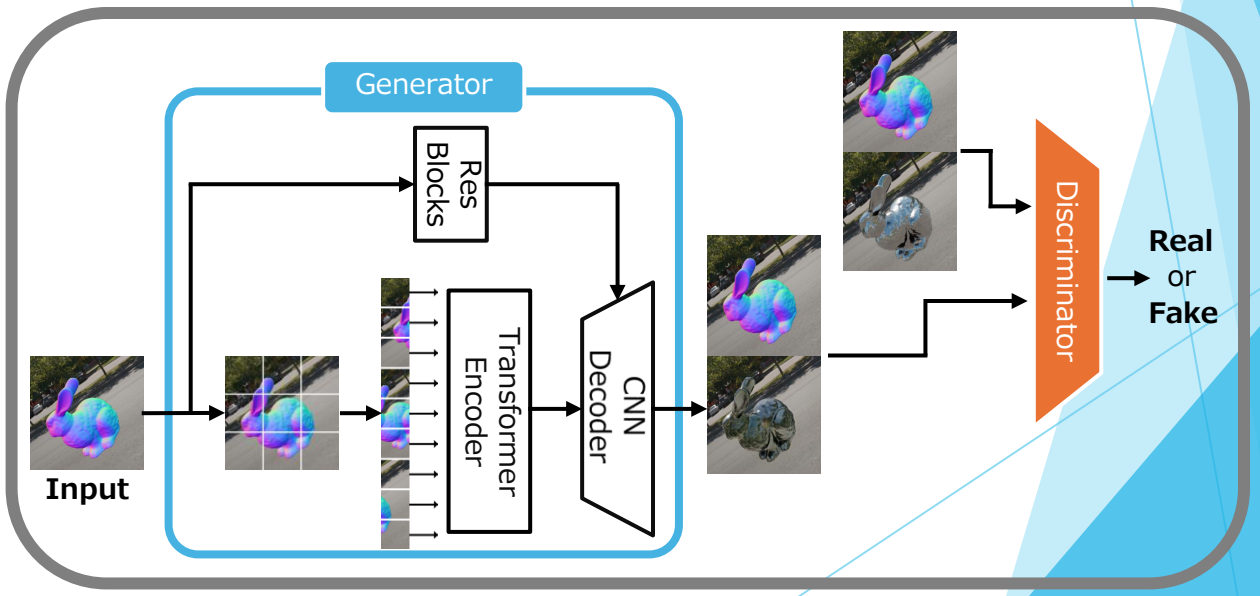
- We introduce ViT, which can capture long-range dependencies in the input image, into the network to more accurately reflect the lighting information in the output image.

Proposed Method

- We adopt pix2pix, a conditional GAN for image-to-image translation, as our framework.
- We introduce ViT into the generator encoder instead of U-Net.
- ViT learns relationships between patches and effectively extracts features that capture long-range dependencies.

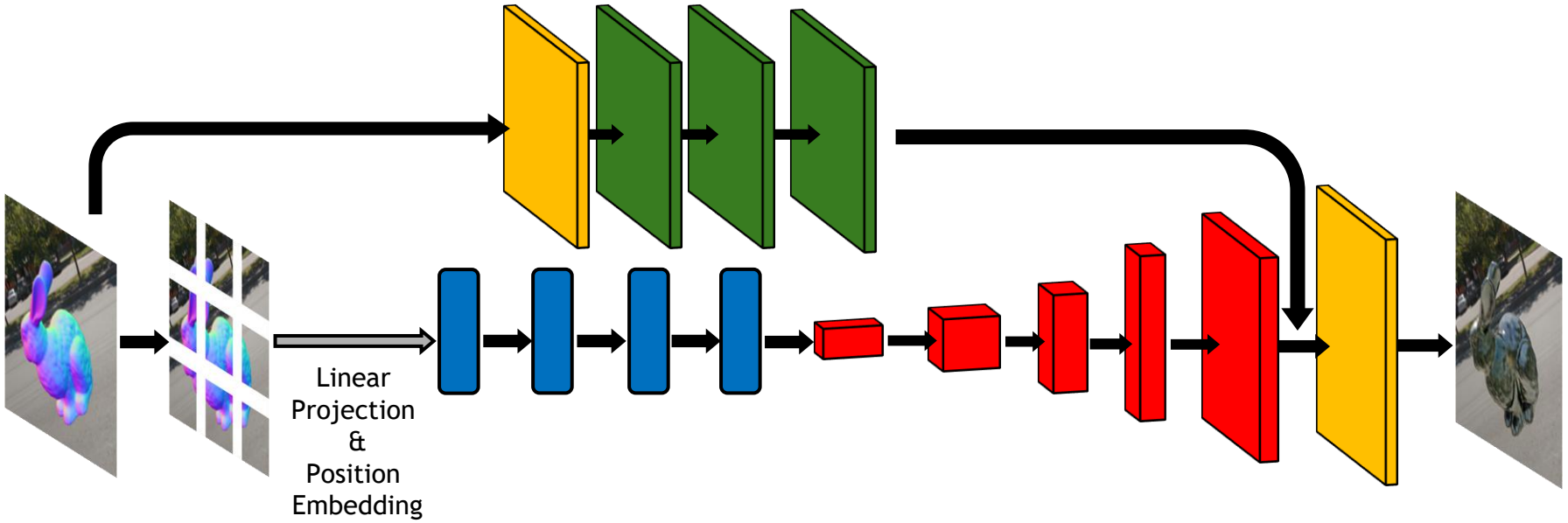


pix2pix

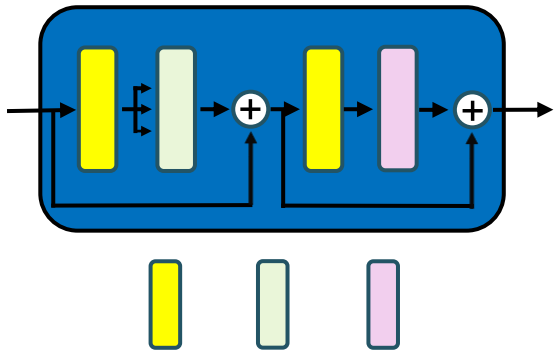


Ours

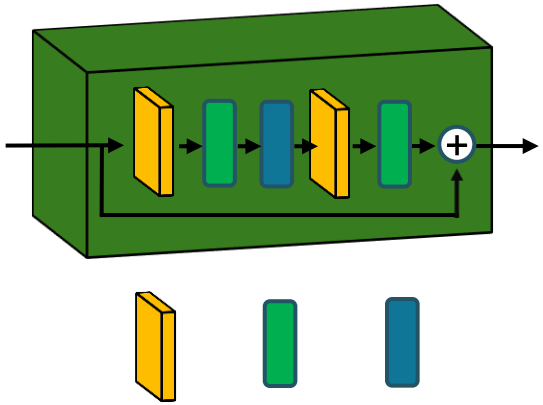
Generator



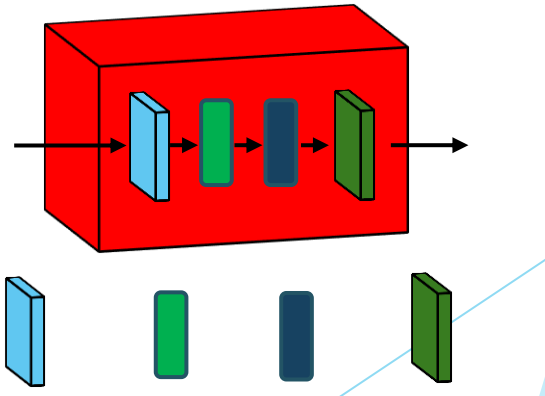
Transformer Block



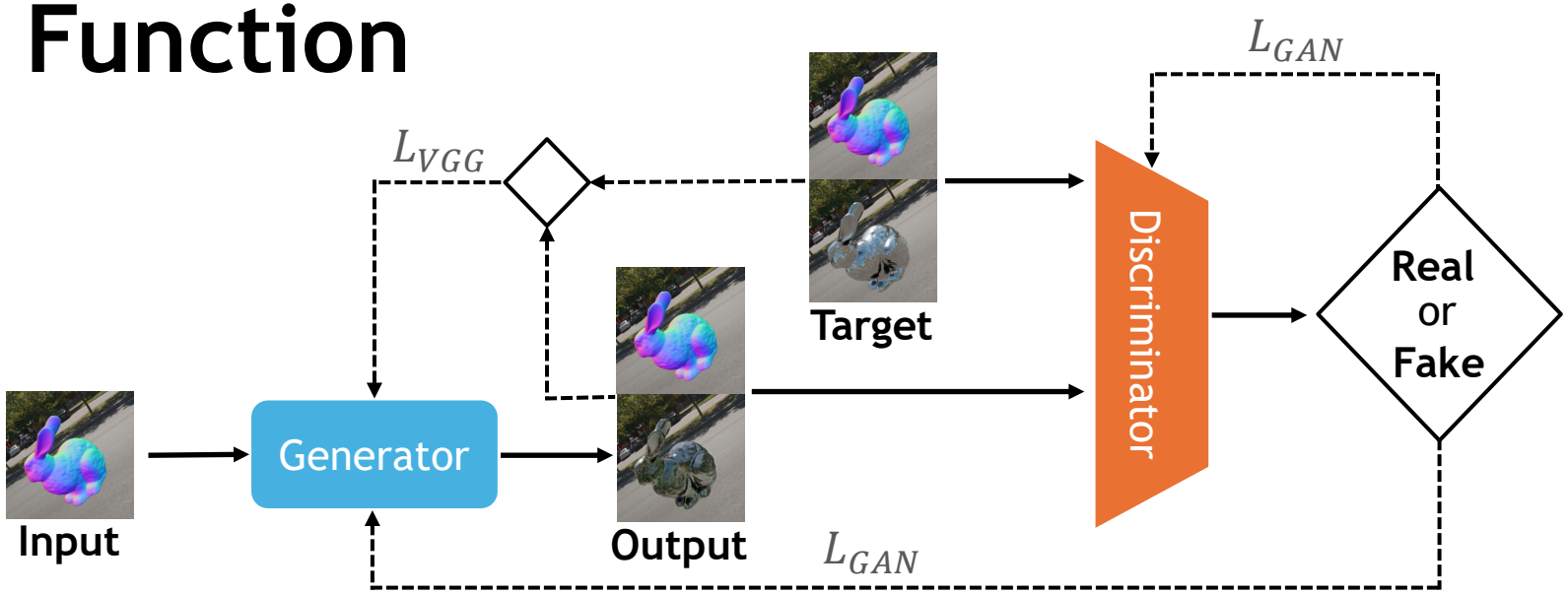
ResBlock



Upsampling Block



Loss Function



- Adversarial Loss (L_{GAN}):

$$L_{GAN}(G,D) = E_{x,y}[\log D(x,y)] + E_{x,G(x)}[\log\left(1 - D\left(x, G(x)\right)\right)]$$

- Perceptual Loss (L_{VGG}):

$$L_{VGG} = \Sigma_j \left\|\varphi_j\left(G(x)\right) - \varphi_j\left(y\right)\right\|$$

$\varphi_j(\cdot)$ is VGG19 intermediate layers

- Overall Loss:

$$L = \min_G \max_D L_{GAN}(G,D) + \lambda L_{VGG}$$

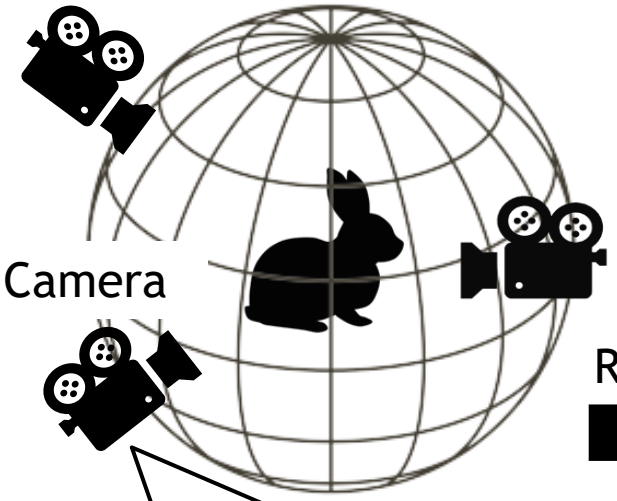
Dataset Preparation

Omnidirectional Image
(High Contrast & Low Contrast)

3D Object

Material Parameter
Metallic: 1.0
roughness: 0.2
specular: 1.0

Environment Map



Camera

Camera Parameter
Radius $r \in [0.25, 0.50]$
Elevation $\theta \in [0, 3/4\,\pi]$
Azimuth $\varphi \in [0, 2\pi]$

Rendering

Input Image

Target Image

Experiment

We compared the proposed method with a baseline method using a CNN-based image-to-image translation network.

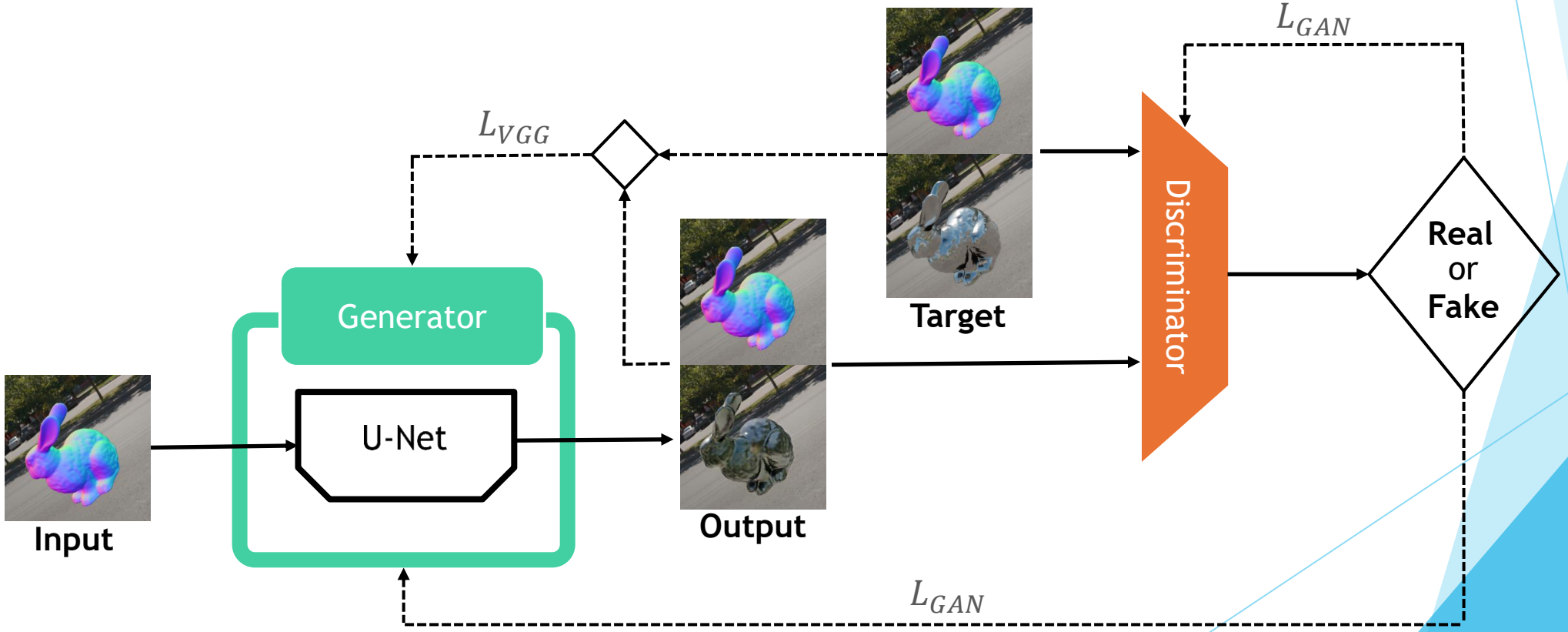
- Image size: 256 × 256
- Training data: 50,000 paired images
- Test data: 10,000 paired images
- Optimization: Adam (β_1 = 0.5, β_2 = 0.999)
- Generator learning rate: 0.0002
- Discriminator learning rate: 0.0001
- Batch size: 16
- Epochs: 100

Training data (example)

Test data (example)

Baseline Method

- The CNN-based image-to-image translation network was pix2pix [Isola et al., 2017].
- The perceptual loss (L_{VGG}) was adapted instead of L_1 loss.



Results (1/3)

Example images taken in a backlit environment.

Input	pix2pix (CNN)	Ours (ViT)	Ground Truth

Results (2/3)

Example images without high-intensity light sources in the background.

Input	pix2pix (CNN)	Ours (ViT)	Ground Truth

Results (3/3)

Example images of failures.

Input	pix2pix (CNN)	Ours (ViT)	Ground Truth

Quantitative Evaluation Results

- PSNR : pixel-wise differences in images.
- SSIM : differences in image structure and contrast.
- LPIPS : perceptual similarity.

	PSNR(↑)	SSIM(↑)	LPIPS(↓)
pix2pix(CNN)	21.28	0.8499	0.1140
Ours(ViT)	**21.67**	**0.8570**	**0.1077**

- SSIM results improved by generating highlights and shading based on the surrounding environment.
- PSNR and LPIPS showed excellent results due to the consistency of the color tone and brightness of the virtual objects.

Discussion

- In addressing the ill-posed task of estimating environments outside the field of view, we prioritized perceptual realism of metallic appearance over pixel-perfect reconstruction.
- The CNN-based method was dependent on local information under backlighting, resulting in overly bright highlights.
- The proposed method captured global lighting context, generating consistent highlights and shading.
- Even when reflected details differed from ground truth, it reproduced the visually coherent appearance of metallic materials and outperformed the CNN-based method in terms of perceptual realism.

Conclusion and Future Work

- Conclusion
  - We proposed a method to reproduce the appearance of metallic materials by using an image-to-image translation network with ViT.
  - The network achieved robust performance under lighting conditions with high-intensity light sources in the background.
- Future Work
  - A network that handles indoor scenes with diverse objects and complex light interreflections.
  - A general-purpose network capable of working in both indoor and outdoor scenes by extracting relevant features from an input image.